KaryoFlow: Few-Step Rectified-Flow Detection for Chromosome Karyotyping

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***Abstract*—Automated chromosome karyotyping requires localizing about 46 densely packed objects from 24 morphologically similar classes in each metaphase image. Small object extent, frequent box overlap, and limited annotated data make pixel-level localization errors and unstable iterative proposals more consequential than in sparse natural scenes. We present *KaryoFlow*, a rectified-flow detector that jointly designs box trajectories and final decisions for this regime. Compared with a denoising diffusion probabilistic detector, its linear noise-to-box formulation increases three-seed mean average precision by 0.0170 and 0.0560 on two public chromosome datasets with different sample sizes and object scales. Direct target-box prediction is algebraically equivalent to velocity prediction but avoids terminal loss amplification, yielding a consistent 0.0020 gain on both datasets. We adapt a second-order multistep solver and show that proposal replacement invalidates its history. Preserving proposal identity maintains three-seed accuracy on both datasets and reduces latency by 2.43%. Finally, a learned localization-quality branch estimates proposal overlap and recalibrates only the final ranking. On a fixed 24 Chromosomes Object checkpoint, it improves mean average precision from 0.8630 to 0.8704; average precision at 90% and 95% intersection-over-union thresholds increases by 0.0304 and 0.0325, respectively, while candidate coordinates, class logits, and trajectories remain fixed before post-processing. Head distillation further reduces latency from 77.57 to 44.72 milliseconds at a 0.0040 precision cost. These results show that aligning trajectory learning, proposal identity, and localization-aware ranking enables accurate and efficient generative detection for dense chromosome analysis.**

***Index Terms*—Rectified Flow, object detection, localization quality, diffusion models, medical image analysis, chromosome karyotyping**

I. Introduction

Chromosome karyotyping remains a labor-intensive step in cytogenetic analysis. Its detection stage is structurally unusual: a metaphase spread contains a nearly fixed set of about 46 chromosomes assigned to 24 classes, many instances occupy less than one percent of the image, and neighboring chromosomes frequently overlap in their bounding-box support. In our two training cohorts, the median instance count is 46; 98.8% and 100% of images contain at least one pair of ground-truth boxes with IoU above 0.1. The median relative box area also shifts from 0.37% to 0.84% between cohorts. Consequently, a few-pixel error can preserve AP50 while changing the strict-IoU ordering that determines mAP, and an iterative detector must refine many adjacent candidates without confusing their identities.

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Conventional detectors offer complementary but incomplete solutions. Anchor-based systems such as Cascade R-CNN [1] and YOLOX [2] are fast, whereas transformer detectors such as DINO [3] obtain strong localization through multi-scale deformable attention [4]. DiffusionDet [5] instead treats boxes as a set generated from noise, an appealing formulation for variable spatial layouts, but its DDPM inheritance introduces a curved stochastic path and a many-step design that are poorly matched to interactive microscopy analysis. More importantly, simply shortening that path does not resolve two detection-specific mismatches: multistep integration assumes a persistent state identity, whereas proposal renewal replaces selected states; and final ranking by class confidence ignores whether a small chromosome is localized accurately enough to survive strict IoU thresholds.

Rectified Flow [6] (RF) offers a natural generation foundation by replacing the DDPM path with a deterministic linear transport between noise boxes and targets. Applying RF to

detection raises three linked questions: how to learn a well-conditioned box trajectory from limited data, how to integrate it without violating proposal identity, and how to rank fixed outputs by both semantic and geometric reliability. KaryoFlow answers them as one trajectory–decision design, shown in Figure 1. Its contributions are threefold.

First, we formulate RF directly in normalized four-dimensional box space and derive a data-prediction objective for the reverse path. Although data and velocity prediction are algebraically equivalent, their squared losses differ by t^{-2}; direct box prediction therefore avoids amplifying errors near the terminal state emphasized by the shifted schedule. The resulting direction is replicated across three seeds on both chromosome datasets. RF itself is adopted, and our claim is the detection formulation and analysis rather than invention of the underlying flow framework.

Second, we derive an RF data-prediction update from DPM-Solver++ [7] and establish the proposal-identity condition for multistep box integration. If a row is renewed between two evaluations, the history correction becomes a secant between different candidates and is not a derivative along either trajectory. Disabling renewal when the proposal pool is redundant restores this premise, retains three-seed accuracy on both datasets, and saves 2.43% latency; its failure at $K = 100$ provides an explicit capacity boundary. DPM-Solver++ itself is used for efficiency, not claimed as an accuracy novelty.

Third, we introduce Localization-Quality Calibrated Ranking (LQCR) for the strict-IoU bottleneck of small, crowded chromosomes. From the final proposal feature, LQCR predicts conditional localization quality and ranks detections by $s = pq^2$. It is applied only when outputs are emitted, leaving candidate coordinates, class logits, proposal survival, and the entire RF trajectory unchanged before final post-processing. The resulting fixed-checkpoint gain is therefore attributable to decision-stage ranking.

We evaluate KaryoFlow on two public chromosome datasets containing 2,200 and 5,000 images, with multi-seed controls, same-checkpoint solver tests, test-set evaluation, strict-IoU analysis, and same-hardware latency profiling. The experiments also separate solver network evaluations from cascade-head evaluations and show that head distillation and conditional stepping compress orthogonal sources of inference cost. Standardized coupling controls are null, so stochastic coupling is excluded from the final model and contribution list.

II. Related Work

a) Chromosome and microscopy detection: Chromosome analysis has progressed from model-based joint segmentation and classification in multispectral images [8] to learned type/polarity classification, chromosome enumeration, and generative straightening [9], [10], [11]. These systems establish the value of chromosome-specific structure, but do not study few-step box-set generation. ChromosomeNet [12] demonstrates learned metaphase-image detection on the same benchmark used here, while modern karyotyping pipelines commonly adapt general detectors such as Faster R-CNN [13], Cascade R-CNN, YOLO, or transformers. Related microscopy work addresses sparse cell localization, dense-boundary segmentation, and cross-modality nucleus detection [14], [15], [16]; together, it shows that crowding and acquisition shift are recurring bioimage-analysis problems. KaryoFlow focuses on the preceding detection stage: accurate, efficient localization of all chromosomes before downstream arrangement and diagnosis.

b) Generative detection and few-step integration: DiffusionDet [5] formulates object detection as iterative box denoising. FlowDet [17] and DeFloMat [18] replace DDPM-style dynamics with conditional flow matching or RF, while DiffuBox [19] studies a distinct 3D refinement setting. RF [6] and Flow Matching [20] provide our adopted continuous-time foundation, and DPM-Solver++ [7] supplies the multistep integration principle. Recent work has also explored flow-matching design and optimized priors for cellular-microscopy generation and medical augmentation [21], [22]; these image-generative settings are complementary to our discriminative transport of box sets. Existing studies emphasize the probability path or solver, but do not analyze the interaction between solver history and proposal renewal. KaryoFlow derives the data-prediction update for the RF box path and makes proposal identity an explicit condition of multistep inference. Diffusion in TMI has mainly addressed synthesis, segmentation, or robust classification [23], [24], [25], leaving few-step medical box-set prediction comparatively underexplored.

c) Localization quality in medical detection: Medical detectors must handle small targets, incomplete labels, acquisition shift, and structured false positives. Representative TMI studies address few-shot pulmonary lesions, heterogeneous lesion annotations, and relational transfer across domains [26], [27], [28], while context refinement, weak supervision, and imbalanced learning reduce task-specific false positives [29], [30], [31]. For general detection, IoU-Net predicts localization confidence for proposal selection [32], and GFL and Varifocal-Net combine class and localization quality in dense heads [33], [34]. LQCR inherits the probability-ranking motivation but differs in causal scope: it is applied only after a generative detector has completed all trajectory updates, so solver history, coordinates, predicted classes, and proposal survival are fixed. This controlled boundary is particularly useful in chromosome images, where AP50 is nearly saturated but strict-IoU errors remain concentrated in small and overlapping instances. The emphasis on output reliability is also consistent with evidence that accurate medical networks can remain poorly calibrated [35].

III. Method

A. Box-Space Rectified Flow

a) RF formulation: The conditional probability path is

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \qquad t \in [0,1], \tag{1}$$

where $\mathbf{x}_0$ is the target (GT bbox) and $\mathbf{x}_1$ is the source (Gaussian noise). The corresponding velocity field is constant along the path of (1), $\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$, and the training objective is the flow matching loss in (2)

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[\left\|\mathbf{v}_\theta(\mathbf{x}_t,t) - (\mathbf{x}_1 - \mathbf{x}_0)\right\|^2\right]. \tag{2}$$

The detection state is a set of normalized (c_x, c_y, w, h) vectors. Thus $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_4)$ is a source box, $\mathbf{x}_0$ is a matched target box, and approximately 46 target trajectories coexist in each metaphase image. Image features condition the velocity field but are not diffused.

b) Time conditioning: AdaLN-Zero [36] injects the time embedding into each proposal head through zero-initialized residual modulation. It is retained as a stable conditioning mechanism; controlled ablation does not support treating it as a separate accuracy contribution.

B. Stable Parameterization and Few-Step Integration

a) Heun solver (2nd-order): For a positive reverse increment Δt, Heun's method gives

$$\begin{aligned} \hat{\mathbf{x}}_{t-\Delta t} &= \mathbf{x}_t - \Delta t\,\mathbf{v}_\theta(\mathbf{x}_t,t), \\ \mathbf{x}_{t-\Delta t} &= \mathbf{x}_t - \tfrac{\Delta t}{2}[\mathbf{v}_\theta(\mathbf{x}_t,t) + \mathbf{v}_\theta(\hat{\mathbf{x}}_{t-\Delta t}, t-\Delta t)]. \end{aligned} \tag{3}$$

Four reverse steps require seven network evaluations because the terminal update uses Euler.

b) DPM-Solver++ (2nd-order multistep, 1 NFE/step): We adapt DPM-Solver++ [7] to the RF linear path. The model directly predicts $\mathbf{x}_0$ (the GT box), and the update uses polynomial interpolation of the $\mathbf{x}_0$ history in t space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $d\mathbf{x}/dt = (\mathbf{x} - \mathbf{x}_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $\mathbf{x}_0(t)$ interpolation yields the update in (4)

$$\begin{aligned} \mathbf{x}_{t_{n+1}} &= \tfrac{t_{n+1}}{t_n}\,\mathbf{x}_{t_n} + \left(1 - \tfrac{t_{n+1}}{t_n}\right)\mathbf{x}_0^{(n)} + \varphi_1\,\mathbf{D}_1, \\ \varphi_1 &= t_{n+1}\log\tfrac{t_n}{t_{n+1}} - t_n + t_{n+1},\ \mathbf{D}_1 = \tfrac{\mathbf{x}_0^{(n)} - \mathbf{x}_0^{(n-1)}}{t_n - t_{n-1}}, \end{aligned} \tag{4}$$

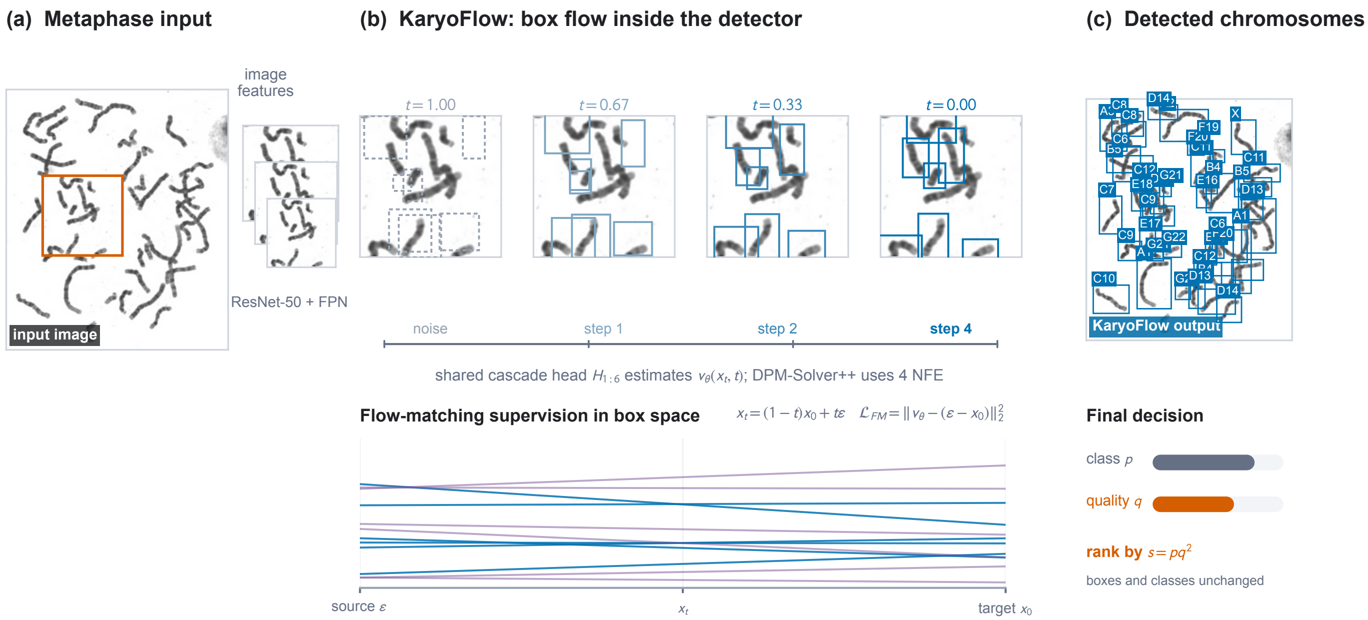

Fig. 1. KaryoFlow from image to detections. A real metaphase image and its dense crop illustrate the feature pyramid, the rectified box path, four-NFE DPM-Solver++ integration, and the final ranked detections. The endpoints and output boxes are empirical; intermediate boxes visualize the prescribed linear RF path rather than recorded hidden activations. Vector overlays remain editable while the microscopy panels retain the source pixels.

where the first two terms are the exact solution for constant $\mathbf{x}_0$ and $\varphi_1\mathbf{D}_1$ is the 2nd-order correction for linearly varying $\mathbf{x}_0(t)$. The singularity at $t\to 0$ is handled by an ϵ cutoff ($t_{n+1} > 10^{-7}$); a 3rd-order variant adds a quadratic term $\varphi_2\mathbf{D}_2$. Unlike VP-SDE DPM-Solver++, $\mathbf{x}_1$ (the initial noise) is a fixed sample and is *not* interpolated; its contribution is carried implicitly by $\mathbf{x}_{t_n}$. *Cost*: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun's 7).

c) Why direct data prediction is numerically preferable here: For the path in (1), $\mathbf{x}_0 = \mathbf{x}_t - t\mathbf{v}$. Hence the two parameterizations carry the same information, but their pointwise squared errors satisfy

$$\mathcal{L}_v(t) = t^{-2}\mathcal{L}_{x_0}(t). \tag{5}$$

With the shifted schedule, velocity regression therefore amplifies gradient noise as $t \to 0$, where accurate terminal boxes matter most. Direct x_0 prediction removes this avoidable conditioning factor. The predicted $+0.002$ validation advantage is reproduced by three seeds on each dataset; test differences remain within noise, so the result supports stable optimization rather than universal statistical dominance.

d) Solver type is mAP*-neutral on RF+Heun:* Evaluating the RF+Heun checkpoint across all solver$\times$step combinations (Table I), solver type has *no effect on* mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect ($+0.004$ from 1 to 4 steps). Heun's $+0.001$ over Euler 4-step costs $1.75\times$ NFE (7 vs 4), not cost-effective. A second same-checkpoint check on the RF-DPM++ model gives 0.863 for four-step DPM-Solver++ and 0.864 for four-step Heun ($\Delta = -0.001$). The solver is therefore an efficiency component rather than an accuracy component.

e) Conclusion: DPM-Solver++ is an efficiency component. On the same RF-DPM++ checkpoint at four steps, DPM-Solver++ and Heun differ by only -0.001 aggregate mAP, while DPM-Solver++ uses four rather than seven network evaluations. The previously observed $+0.006$ cross-checkpoint difference compared models trained under different configurations and cannot be attributed to the inference solver alone. We therefore claim matched accuracy at lower NFE, not a solver-induced accuracy improvement.

TABLE I
SOLVER$\times$STEP DISENTANGLEMENT ABLATION ON THE RF+HEUN CHECKPOINT (DATASET 2 VAL, SEED 42). SOLVER TYPE IS mAP-NEUTRAL AT MATCHED STEPS, WHILE INCREASING ONE TO FOUR STEPS ADDS ONLY 0.004 MAP.

Solver	Steps	NFE	mAP
Heun	4	7	0.8560
Euler	4	4	0.8550
DPM-Solver++	4	4	0.8550
Euler	1	1	0.8510
DPM-Solver++	1	1	0.8510

C. History-Consistent Proposal Integration

a) Proposal identity is a premise of multistep history: Let $r_i^n = 1$ indicate that proposal row i is renewed after step n.

Proposition 1 (Identity consistency)**.** *The multistep correction* $\mathbf{D}_{1,i}$ *in* (4) *is a consistent first-order approximation to a trajectory derivative only if the two states indexed by row* i *belong to the same proposal trajectory, or if the row-wise history is reset after replacement.*

Proof. Without renewal, differentiability of $\mathbf{x}_{0,i}(t)$ gives the standard secant limit

$$\mathbf{D}_{1,i} = \frac{\mathbf{x}_{0,i}^{(n)} - \mathbf{x}_{0,i}^{(n-1)}}{t_n - t_{n-1}} \longrightarrow \frac{d\mathbf{x}_{0,i}}{dt}. \tag{6}$$

If $r_i^n = 1$, the numerator subtracts predictions from two independently initialized states. Its limit contains a finite cross-trajectory jump divided by $t_n - t_{n-1}$ and therefore is not the derivative along either trajectory. Resetting history removes the invalid correction. □

We use the simpler alternative of disabling renewal after verifying sufficient proposal redundancy. This choice is well matched to metaphase images: the object count is tightly concentrated near 46, so $K = 500$ provides multiple candidates per chromosome, while maintaining identity avoids mixing adjacent, morphologically similar instances. The $K = 100$ counterexample, where mAP drops by 0.0313, shows that history consistency cannot compensate for inadequate proposal capacity.

D. Orthogonal Inference Compression

a) Cascade-head distillation: Solver NFE counts network calls, whereas each call evaluates H cascade heads. We distill a six-head teacher into a three-head student using head mapping $\{0\to 0, 1\to 2, 2\to 5\}$ and feature/prediction supervision. At four solver steps this reduces cascade-head evaluations (CHE) from 24 to 12, while NFE remains four; the reduction is therefore 24-to-12 CHE, not NFE. The compressed model reaches 0.859 versus 0.863 and reduces RTX A6000 latency from 77.57 to 44.72 ms.

b) Conditional computation: Geometry-aware adaptive conditional stepping (GACS) freezes convergence and confidence thresholds on seed 42, then chooses one or two steps per image. It is an optional speed policy rather than the accuracy model: on Dataset 1 it uses 1.6136 mean steps, exits early on 38.64% of images, and reduces latency by 13.79% at -0.0007 mAP versus fixed two-step inference. Its mAP remains below the four-step reference, so we do not call it lossless relative to the production setting.

E. Localization-Quality Calibrated Ranking

For prediction feature F, class-correctness event C, localization IoU U, and evaluation threshold τ, the true-positive posterior factorizes exactly:

$$P(C{=}1, U{\geq}\tau \mid F) = P(C{=}1 \mid F)P(U{\geq}\tau \mid C{=}1, F). \quad (7)$$

The probability-ranking principle therefore favors ordering detections by the joint class–localization posterior, rather than class confidence alone. The quality branch predicts $q = \sigma(g_\phi(F)) \in (0,1)$ from the final proposal feature. It is attached only to the sixth (last) cascade head and is implemented as

$$g_\phi(F) = W_2\,\mathrm{ReLU}(\mathrm{LN}(W_1 F)) + b_2, \qquad W_1 \in \mathbb{R}^{128\times 256}, \quad (8)$$

adding only 33,153 trainable parameters. The backbone, feature pyramid, proposal interaction layers, class head, and box head are frozen while this branch is fitted; features entering the branch-only training path are detached.

Proposition 2 (Threshold-consistent quality statistic)**.** *Suppose $P(U \mid C{=}1, F)$ belongs to a one-parameter family that is first-order stochastically ordered by its mean, and q is calibrated to that conditional mean. Then $q_a \geq q_b$ implies $P(U_a \geq \tau \mid C_a{=}1, F_a) \geq P(U_b \geq \tau \mid C_b{=}1, F_b)$ for every threshold τ.*

Proof. First-order stochastic dominance orders every increasing expectation, including the mean and the indicator $\mathbf{1}\{U \geq \tau\}$. Calibration therefore makes q an order-preserving statistic for each survival term. □

The proposition describes the ideal conditional-localization statistic. Our trainable surrogate also assigns zero to unmatched proposals. If M denotes Hungarian matching, its population target is therefore $\mathbb{E}[MU \mid F]$, not a perfectly calibrated $P(U \geq \tau \mid C, F)$. On the matched-candidate manifold it regresses IoU; off that manifold it additionally suppresses background proposals. Thus (7) motivates the ordering, while the exponent sweep below empirically tests whether this practical surrogate improves COCO ranking.

For Hungarian-matched proposal i, let $m_i = 1$ and let u_i be the detached aligned IoU; unmatched proposals have $m_i = 0$. We train the scalar logit with the continuous target $y_i = m_i u_i$ and the varifocal-style objective

$$\begin{aligned} \mathcal{L}_q &= \frac{\lambda_q}{\max(N_+, 1)} \sum_i w_i\,\ell_{\mathrm{BCE}}(g_i, y_i), \\ w_i &= m_i u_i + (1 - m_i)\alpha q_i^\gamma, \end{aligned} \quad (9)$$

where ℓ_{BCE} is binary cross-entropy with logits, $\lambda_q = 0.25$, $\alpha = 0.75$, and $\gamma = 2$. Detaching u_i prevents this auxiliary ranking loss from changing box geometry. We use $s = pq^2$ as a low-cost cross-threshold surrogate for the product in (7). The exponent controls the relative influence of localization quality. A learned-quality sweep over $\beta \in \{0, 0.25, 0.5, 1, 2\}$ selects $\beta = 2$ on validation data; importantly, this is distinct from the true-IoU oracle sweep used only to measure headroom. The proposition justifies the quality ordering, not universal optimality of the exponent for arbitrary IoU distributions.

Our strict final-only implementation keeps raw class scores in the solver, proposal renewal, Top-K pruning, and early stopping. Calibration is applied only when emitting detections, so boxes, predicted classes, and RF trajectories remain fixed at the candidate level; score-dependent NMS or max-detection selection may consequently choose a different final subset. An elementwise checkpoint audit confirms that all 590 tensors shared with the A4 detector are identical; the LQCR checkpoint contains only the five additional tensors of (8). This design targets the observed chromosome regime in which AP50 is nearly saturated but small and overlapping instances lose recall at IoU 0.90. Figure 2 summarizes the principle, causal boundary, and fixed-seed controlled effect.

F. Coupling Audit

We evaluated Random, Hard-OT, and Sinkhorn-sampled coupling under the same augmentation and evaluation protocol. Although deterministic assignment can reduce pairing diversity in the four-dimensional box space, the controlled multi-seed experiments do not show a reliable mAP advantage for stochastic coupling on either chromosome dataset.

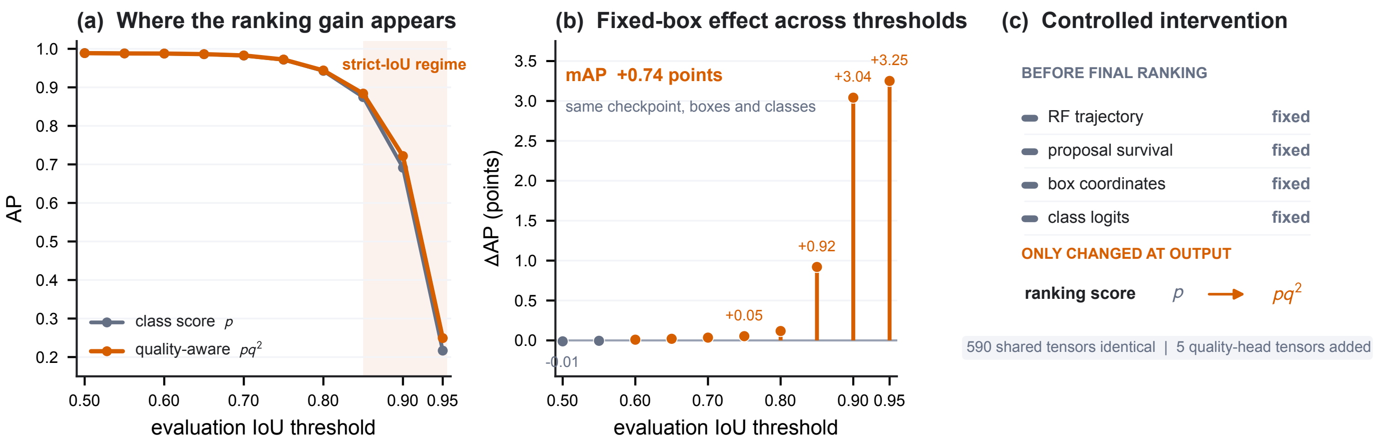

Fig. 2. Effect and causal scope of localization-quality ranking. (a,b) On the fixed Dataset 2 seed-42 RF-DPM++ checkpoint, the gain is concentrated at strict IoU thresholds (AP90/AP95 $+3.04/+3.25$ points), while AP50 is unchanged. (c) The intervention changes only final ordering from p to pq^2; the RF trajectory, renewal, candidate coordinates, and class logits are fixed. Score-dependent post-processing may select a different final subset. Values are loaded from checkpoint-linked exports.

This audit is motivated by conditional and multisample flow matching [37], [38] and uses entropic Sinkhorn transport [39]; it also agrees with evidence that conditional OT can introduce its own failure modes [40]. Coupling is therefore treated as an auxiliary design choice and negative-result analysis, not as a component of the final method or an accuracy contribution. The complete model uses Random coupling to remove this confound.

G. Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top-K proposals proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch ($500 \to K$).

IV. Experiments

A. Experimental Setup

a) Datasets: Table II summarizes the two public chromosome datasets, referred to as Dataset 1 and Dataset 2. Dataset 1 is Chromosome20240904 (RST) [41], with 2,200 images; Dataset 2 is the 24 Chromosomes Object dataset [42], [43], with 5,000 images. Their training splits contain 1,540 and 3,500 images, respectively. Both have a median of 46 annotated instances per image, but their median relative box areas differ substantially (0.37% versus 0.84%). Moreover, 98.8% and 100% of training images contain at least one ground-truth box pair with IoU above 0.1. These statistics quantify the small, crowded regime and the scale shift used for cross-dataset replication. Both datasets use image-level random splitting. A single blood sample may yield multiple metaphase images, but patient identifiers are unavailable; thus patient-level separation cannot be verified. We analyze only the publicly released images and annotations and collect no additional human-subject data.

TABLE II
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

TABLE III
TRAINING-SET PROPERTIES THAT MOTIVATE THE DETECTOR DESIGN. RELATIVE AREA IS BOX AREA DIVIDED BY IMAGE AREA; OVERLAP DENOTES AN IMAGE CONTAINING AT LEAST ONE PAIR OF GROUND-TRUTH BOXES WITH IOU > 0.1.

Dataset	Mean n	Median n	Med. rel. area	Overlap
Dataset 1	46.77	46	0.37%	98.8%
Dataset 2	45.97	46	0.84%	100.0%

b) Architecture and training: Our base framework (KaryoFlow) uses ResNet-50 [44] + FPN [45] (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [46] (lr$=5\times10^{-5}$, wd$=10^{-4}$), 5-epoch warmup + CosineAnnealing [47], 150 epochs. Loss: Focal [48] (λ_{cls}=2.0) + L1 (λ=5.0) + GIoU [49] (λ=2.0) with Hungarian matching [50]. Diffusion uses RF with shifted noise schedule (shift$=3.0$); default solver is DPM-Solver++ 4-step. For the isolated LQCR intervention, the trained A4 checkpoint initializes the model, all pre-existing parameters are frozen, and only the 33k-parameter quality branch is optimized for at most 12 epochs with AdamW learning rate 10^{-3} and cosine decay; the epoch-2 validation checkpoint is used. Thus this phase fits a ranking statistic but cannot alter the detector's generated boxes or class logits.

c) Detection protocol: Boxes are represented in two spaces: image space (absolute pixel xyxy) for RoIAlign and NMS, and diffusion space (cxcywh normalized to $[0,1]$, mapped to $[-s,+s]$, s=2.0). The RF forward path is $x_t = (1-t)\,x_0 + t\,\varepsilon$ (DDPM baseline uses cosine schedule). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble, predictions from all steps are concatenated and deduplicated by per-class NMS (IoU 0.5). Evaluation uses COCO mAP@0.5:0.95 (10 IoU thresholds, maxDets$=100$).

d) Statistical considerations: Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, the DPM-Solver++ variant has 3 seeds (mean 0.859 ± 0.003), which sets the scale for interpreting small differences. The $+0.006$ cross-checkpoint solver difference compares independently trained models and is not a solver-causal result; the same-checkpoint comparison is neutral. LQCR is instead evaluated by a fixed-checkpoint, final-only intervention, so its ranking effect is isolated from training and trajectory changes.

e) Evidence traceability: Every completed numerical claim is registered in a project-level SQLite evidence database with dataset, split, seed, checkpoint, protocol, source path, and SHA-256. Controlled results are separated from running snapshots and diagnostic oracles. For LQCR, the registry records both checkpoint hashes, the elementwise identity audit of all 590 shared tensors, five learned-q sweep exports, and a separate label identifying true-IoU oracle fields. Dataset-property statistics likewise store hashes of the two raw training annotation files. This layer does not replace independent replication, but prevents stale plots, cross-checkpoint comparisons, and oracle quantities from silently entering the reported evidence.

B. Precision Bottleneck Diagnosis

Figure 3 separates classification, ranking, and coordinate headroom before proposing further modules. Perfect-class oracles add little compared with perfect-coordinate oracles, and AP degrades rapidly at strict IoU thresholds. At IoU 0.90, small and overlapping chromosomes show the largest recall deficits. These observations identify localization accuracy and localization-aware ordering—rather than class recognition alone—as the current precision bottleneck.

C. Main Results: RF vs DDPM

a) Dataset 2 ablation: Table IV reports a cumulative ablation on the Dataset 2 validation set. The historical DDPM and RF+Heun rows establish the adopted generation foundation. The same seed-42 RF-DPM++ checkpoint then isolates the four-NFE DPM-Solver++ operating point and the final-only LQCR intervention. AdaLN-Zero is used throughout as time conditioning but is not counted as an independent contribution.

RF supplies the dominant historical improvement over the DDPM baseline. DPM-Solver++ retains accuracy while reducing Heun's seven evaluations to four; same-checkpoint solver tests do not support an additional precision claim. LQCR then improves mAP by 0.0074 without changing the checkpoint, candidate coordinates, class logits, renewal decisions, or trajectory. Its largest gains occur at strict IoU thresholds (AP90/AP95 $+0.0304/+0.0325$), consistent with the localization-aware ranking mechanism rather than a coordinate change.

These interventions were not selected from isolated best checkpoints. Renewal, parameterization, and coupling conclusions use multi-seed controls; LQCR uses a deliberately fixed checkpoint and final-only causal boundary; GACS freezes thresholds before cross-seed evaluation; and latency is measured on one RTX A6000. Failed and null directions remain in the same database, preventing selective omission from being mistaken for a complete search history.

b) Dataset 1: RF vs DDPM: On Dataset 1 (3 seeds), RF outperforms DDPM by $+0.017$ mAP (0.746 vs 0.729, lower variance ± 0.001 vs ± 0.004) with 4-step inference vs DDPM's 1-step. DDPM gains only $+0.044$ from $1 \rightarrow 8$ steps ($0.628 \rightarrow 0.672$), whereas the RF paradigm yields $+0.082$ mAP on Dataset 2.

D. SOTA Comparison

Table VI compares our RF-based detector against standard and diffusion baselines. With a standard ResNet-50–FPN backbone, KaryoFlow enters the same high-accuracy group as the stronger CSPNeXt-L RTMDet [51] and multi-scale DINO R50 [3], while exceeding Cascade R-CNN, YOLOX-S [2], and the DDPM-based DiffusionDet [5]. The largest controlled benchmark separation is from DiffusionDet: $+0.060$ mAP for the seed-42 checkpoint and $+0.056$ for the KaryoFlow three-seed mean. Relative to RTMDet-L and DINO R50, the remaining gaps are 0.004 and 0.009 mAP, respectively. KaryoFlow simultaneously uses $1.71\times$ fewer network evaluations than RF+Heun, showing that generative box prediction can be accurate without retaining a many-step diffusion sampler.

a) Small-object performance: On small objects, our detector (AP_S=0.516 over three seeds) is close to DINO R50 (0.553) and RTMDet-L (0.540), but the validation split contains only 60 images with small objects and the point estimates are high-variance. We therefore do not claim a small-object *advantage*; rather, a single-shot RF detector with a plain ResNet-50 backbone is *competitive* on the small-object regime most relevant for chromosome analysis, without the multi-scale deformable attention of DINO or the heavier CSPNeXt-L backbone of RTMDet-L. On the smaller Dataset 1 training split ($1{,}540$ images), the KaryoFlow three-seed mean is 0.747 mAP (best checkpoint 0.753), compared with 0.742 for both RTMDet-L and DINO R50. This second cohort supports competitiveness under limited data, while the larger Dataset 2 comparison shows that the advantage is not universal across all detector families.

b) Class- and scale-resolved analysis: The diagnostic exports show that residual errors are structured rather than uniform. High-IoU recall falls most for small chromosomes and locally overlapping instances (Figure 3), matching the training-set statistics in Section IV-A. At the conventional IoU 0.50 threshold, performance is already near saturation; the remaining headroom therefore lies in sub-pixel coordinate refinement and quality-aware ordering, not simply in producing more low-quality proposals. This observation motivates LQCR and explains why its gain concentrates at AP90/AP95.

c) Causal interpretation of controlled comparisons: The same-checkpoint solver comparison is accuracy-neutral, and

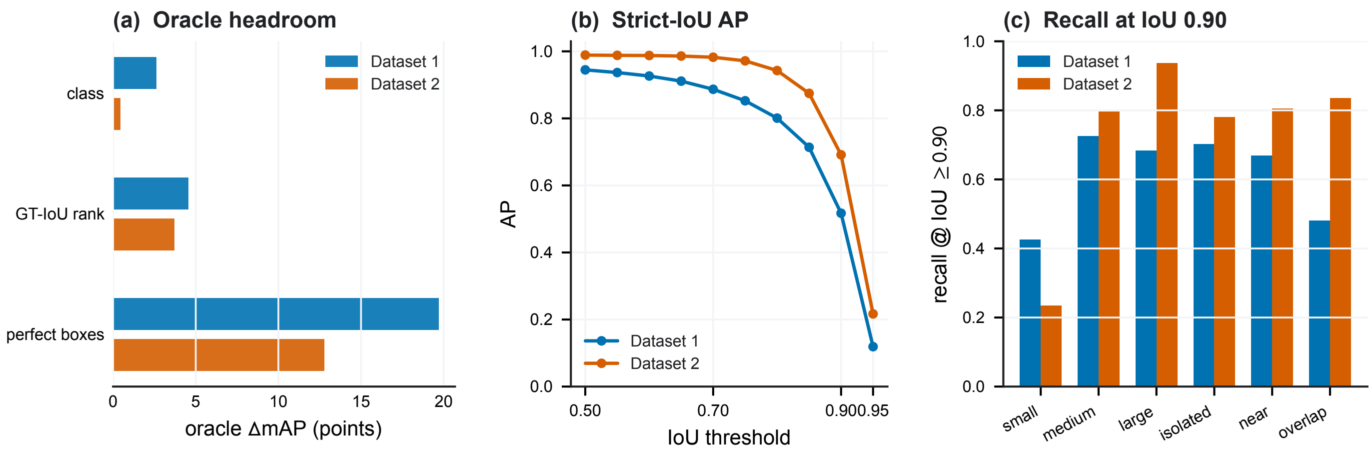

Fig. 3. Precision bottleneck diagnosis. (a) Oracle mAP headroom for perfect classes, ground-truth-IoU ranking, and perfect coordinates. Ground-truth-IoU ranking is a diagnostic upper bound, not a trained LQCR result. (b) AP versus IoU threshold. (c) Detection recall at IoU 0.90 by scale and overlap condition. Dataset 1 uses a seed-42 100-image diagnostic subset; Dataset 2 uses the seed-42 500-image validation export.

TABLE IV
EVIDENCE-CONTROLLED ABLATION ON DATASET 2 VALIDATION. THE FINAL TWO ROWS SHARE THE SEED-42 RF-DPM++ CHECKPOINT; LQCR CHANGES ONLY OUTPUT RANKING.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP$_S$	AP$_M$	AP$_L$
DDPM baseline (Euler 1-step)	Euler	1	1	0.7740	0.9680	0.9160	0.3170	0.7730	0.7750
RF+Heun (KaryoFlow)	Heun	4	7	**0.8560**	0.9890	0.9690	0.5020	0.8530	0.8670
DPM-Solver++ (seed 42)	DPM++	4	4	0.8630	0.9889	0.9717	0.5315	0.8594	0.8921
+LQCR (final-only)	DPM++	4	4	**0.8704**	0.9888	0.9722	0.5304	0.8666	0.9142

TABLE V
CONTROLLED INDEPENDENT AND ENGINEERING INTERVENTIONS. MEANS USE THREE SEEDS UNLESS MARKED SEED 42. CHE DENOTES CASCADE-HEAD EVALUATIONS AND IS DISTINCT FROM SOLVER NFE. EVERY ROW IS STORED WITH SOURCE PATH, PROTOCOL, AND PAPER-ELIGIBILITY STATUS IN `TOOLS/EXPERIMENT_DB/EXPERIMENTS.DB`.

Intervention	Controlled comparison	Accuracy effect	Efficiency effect	Interpretation / boundary
LQCR	D2, fixed checkpoint, seed 42	$+0.0074$ mAP	negligible	Final ranking only; candidate states fixed before post-processing
Renewal off	D2 / D1, $K = 500$	$-0.0003/-0.0030$	-2.43% latency	Restores DPM++ history identity; fails at $K = 100$ (-0.0313)
Head distillation	D2, H6 teacher vs H3 student	-0.004 mAP	$1.73\times$; CHE $24 \rightarrow 12$	NFE remains four; D1 H3S4 is architecture-only control
GACS	D1, adaptive vs fixed two-step	-0.0007 mAP	-13.79% latency	Not lossless versus four-step reference (0.747)
x_0 prediction	D1 / D2, 3-seed vs velocity	$+0.002/+0.002$ mAP	no added cost	Matches t^{-2} conditioning analysis; test difference is noise-scale

TABLE VI
DETECTOR COMPARISON ON DATASET 2. KARYOFLOW RESULTS REPORT THE THREE-SEED MEAN; ALL METHODS USE INDEPENDENTLY TRAINED CHECKPOINTS.

Method	Backbone	mAP	AP50	AP75	AP$_S$
DINO R50	ResNet-50	**0.8680**	0.9920	0.9790	0.5530
RTMDet-L	CSPNeXt-L	0.8630	0.9920	0.9760	0.5400
Ours (DPM-Solver++)	ResNet-50	0.8590	0.9880	0.9680	0.5160
Ours (RF+Heun)	ResNet-50	0.8560	0.9890	0.9690	0.5020
Cascade R-CNN	ResNet-50	0.8540	0.9870	0.9720	0.5250
YOLOX-S	CSPDarkNet-S	0.7960	0.9870	0.9440	0.4520
DiffusionDet	ResNet-50	0.8030	0.9700	0.9280	0.5000

standardized multi-seed coupling experiments do not establish an accuracy contribution. The retained positive intervention is LQCR: enabling its score only at the final output changes mAP from 0.8630 to 0.8704 on exactly the same checkpoint and detections before ranking. This paired construction isolates decision-stage calibration from generation, interaction, and numerical integration effects. We report this as a fixed-checkpoint causal intervention, not as a cross-seed training gain; the independent multi-seed evidence in the paper concerns RF, prediction parameterization, renewal, and coupling.

The monotone mAP trend supports using $\beta = 2$, while invariant AP50 shows that calibration reorders already detected candidates rather than creating new easy true positives. AP90 and AP95 rise at every point in the sweep, matching the threshold-factorization prediction. Full-precision values and

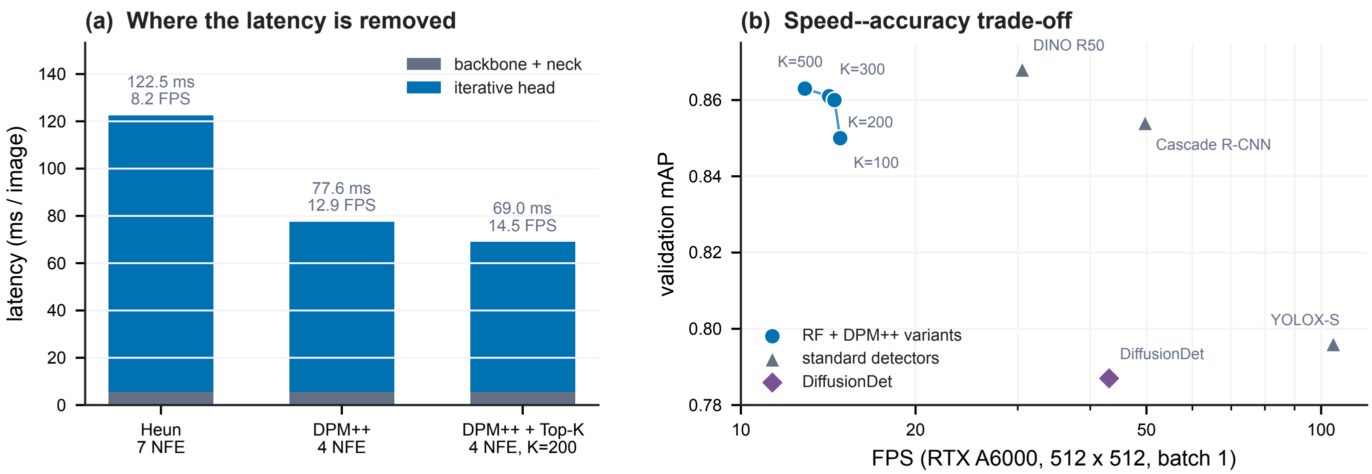

Fig. 4. Same-hardware efficiency on Dataset 2. (a) Latency decomposition. (b) Validation mAP versus FPS on one RTX A6000; connected points form a checkpoint-paired Top-K sweep, whereas other detectors are independent.

TABLE VII
LEARNED-QUALITY EXPONENT SWEEP ON DATASET 2 VALIDATION (SEED 42). THE $\beta = 0$ ROW DISABLES QUALITY RANKING. THESE ARE LEARNED-q RESULTS, NOT GROUND-TRUTH-IOU ORACLE RANKINGS; ALL ROWS USE THE SAME SINGLE-IMAGE PRECISION-DIAGNOSTIC PROTOCOL.

β	mAP	ΔmAP	AP50	AP75	AP90	AP95
0	0.8630	—	0.9889	0.9717	0.6914	0.2167
0.25	0.8657	+0.0027	0.9889	0.9718	0.7022	0.2284
0.5	0.8673	+0.0043	0.9889	0.9719	0.7086	0.2352
1	0.8689	+0.0059	0.9888	0.9717	0.7158	0.2422
2	**0.8704**	**+0.0074**	0.9888	0.9722	**0.7218**	**0.2492**

TABLE VIII
SAME-HARDWARE LATENCY BENCHMARK ON DATASET 2 AT 512×512 RESOLUTION (RTX A6000, BATCH SIZE 1, 500 TIMED ITERATIONS).

Model	Solver	NFE	Latency (ms)	FPS	mAP
RF+Heun	Heun	7	122.55	8.2	0.856
DPM-Solver++	DPM++	4	**77.57**	**12.9**	**0.863**
DPM-Solver++ +Top-K (K=300)	DPM++	4	70.46	14.2	0.861
DPM-Solver++ +Top-K (K=200)	DPM++	4	**68.99**	**14.5**	**0.860**
DPM-Solver++ +Top-K (K=100)	DPM++	4	67.43	14.8	0.850
Cascade R-CNN	—	1	20.11	49.7	0.854
YOLOX-S	—	1	9.53	105.0	0.796
DiffusionDet	Euler	1	23.19	43.1	0.787

the five raw diagnostic exports are checksum-linked in the evidence registry.

1) Qualitative Comparison: Figure 5 compares three localization-sensitive validation cases: a rare Y chromosome, adjacent small G-group chromosomes, and a locally overlapped D-group chromosome. The selected examples show broadly similar localization among KaryoFlow, DiffusionDet, and DINO R50; quantitative claims therefore rely on the complete-set metrics rather than selected images.

E. Controlled Coupling Result

With standardized augmentation, the three coupling strategies are equivalent within cross-seed variation on both datasets. On Dataset 1, their three-seed means lie within approximately 0.002 mAP; on Dataset 2, the stochastic-minus-random difference is also non-significant. Earlier larger gains pooled incompatible experimental lineages and are superseded. We retain this result to document the falsified direction, while excluding Stochastic Coupling from the model definition and contribution list.

F. Solver Analysis

a) DPM-Solver++ step ablation: DPM-Solver++ converges at 2 steps (mAP 0.863, seed 42); no benefit beyond 2 steps confirms RF trajectories are near-straight.

b) DPM-Solver++ vs Heun at matched NFE: At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) $\approx$ Heun 2-step (3 NFE, 0.863) — equal accuracy, so DPM-Solver++ achieves 43% fewer NFE at no cost. At matched *step count* on the same RF-DPM++ checkpoint, DPM-Solver++ gives 0.863 and Heun gives 0.864 ($\Delta = -0.001$), which is practically neutral. We therefore attribute no accuracy gain to the off-the-shelf solver; its value is the lower evaluation count and latency.

c) Test set evaluation: On the Dataset 2 test set, DPM-Solver++ achieves mAP 0.859 (vs val 0.863 for seed 42, $\Delta = -0.004$), confirming that the aggregate accuracy generalizes well. The AP_S point estimate, however, swings from 0.499 (val) to 0.577 (test) for the same checkpoint, indicating that AP_S on this 24-class benchmark is high-variance (only 60 val images contain small objects). We therefore base the small-object discussion on the full scale- and overlap-resolved diagnostics rather than on one split-specific AP_S point estimate.

G. FPS / Latency Benchmark

Table VIII and Figure 4 report the speed-accuracy trade-off at 512×512 input resolution. DPM-Solver++ with Top-K pruning reaches a latency compatible with interactive use, while single-shot detectors (Cascade R-CNN, YOLOX-S) trade accuracy for speed.

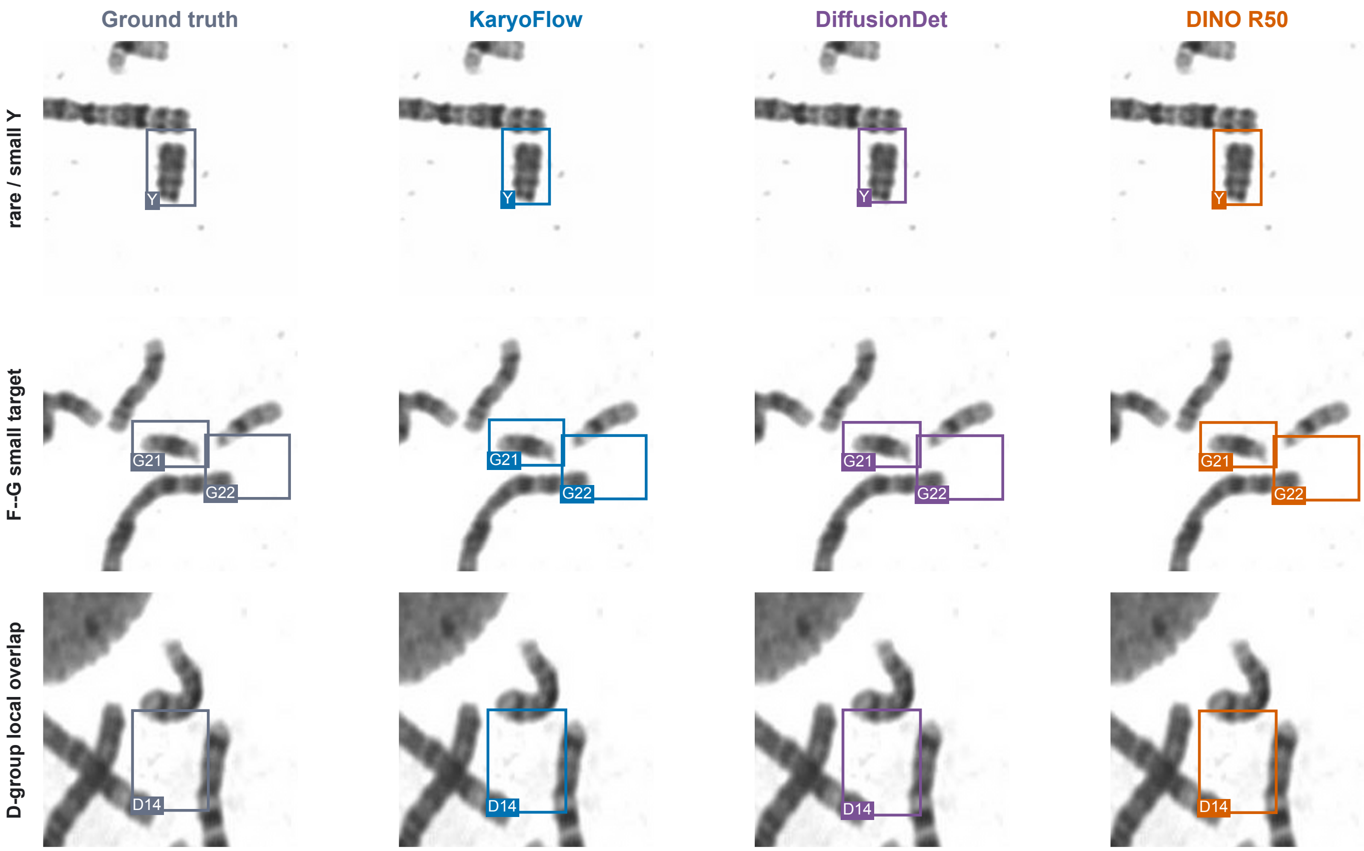

Fig. 5. Local difficult-case comparison on Dataset 2 validation images 1, 2, and 4. Columns show ground truth, KaryoFlow, DiffusionDet, and DINO R50; rows focus on Y, F/G-group, and D-group target categories. Predictions use a score threshold of 0.30. Only the row's target category group is overlaid to keep localization readable; microscopy crops are raster, while boxes, labels, and layout remain vector-editable. These selected cases are illustrative rather than an accuracy estimator.

DPM-Solver++ +Top-K (K=200) reaches $68.99\,\mathrm{ms}$ / 14.5 FPS at $\mathrm{mAP}\ 0.860$; DPM-Solver++ achieves $77.57\,\mathrm{ms}$ / 12.9 FPS at $\mathrm{mAP}\ 0.863$. The cascade head dominates $90\%+$ of latency; the backbone+neck is a minor cost ($\sim 5.8\,\mathrm{ms}$, 4–8%).

The engineering audit separates three counts that are often conflated: DPM-Solver++ reduces solver NFE ($7 \rightarrow 4$), head distillation reduces CHE ($24 \rightarrow 12$ at four NFE), and GACS reduces mean per-image steps ($2 \rightarrow 1.6136$). Renewal removal does not change any count; it makes multistep history coherent and eliminates proposal replacement overhead.

H. Sensitivity to Annotation Precision

To quantify why strict localization matters, we perturb only the Dataset 2 test annotations while keeping images and predictions fixed. A 5-pixel Gaussian center perturbation reduces AP50 by 0.141 but AP75 by 0.596. This is an evaluation-sensitivity diagnostic rather than a model-robustness claim: modest annotation uncertainty is amplified at strict IoU, further motivating AP90/AP95 analysis and localization-aware ranking for small chromosomes.

V. Discussion

KaryoFlow's components address different consequences of metaphase-image structure. The low-dimensional linear path and direct box prediction reduce optimization burden when annotated cohorts are modest. Proposal identity matters because dozens of adjacent candidates evolve simultaneously, while the nearly fixed chromosome count permits a redundant proposal pool that makes renewal unnecessary at the retained operating point. LQCR then addresses the remaining decision bottleneck: class confidence is already sufficient at IoU 0.50, but it does not identify which small or overlapping boxes will survive stricter localization thresholds. The concentration of LQCR gains at AP90 and AP95 is consistent with this task-derived mechanism.

The efficiency results reveal complementary deployment controls rather than a single opaque speedup. DPM-Solver++ reduces solver evaluations, distillation reduces cascade-head evaluations, Top-K reduces proposals after the first step, and GACS reduces mean per-image steps. This decomposition enables an accuracy-oriented setting (0.863 mAP, 77.57 ms) or a compressed setting (0.859 mAP, 44.72 ms) without changing the central trajectory–ranking formulation.

The study has three boundaries. First, both cohorts are public chromosome datasets and lack patient identifiers, so patient-level leakage and clinical workflow outcomes cannot be assessed. Second, LQCR is isolated on one fixed Dataset 2 validation checkpoint, and the same validation cohort selects β and quantifies the intervention. Its within-checkpoint ranking effect is clear, but a locked test evaluation, broader multi-seed replication, and natural-image validation are required to establish generality. Third, history-consistent inference relies on proposal redundancy: the observed $K = 100$ failure should be treated as an operating constraint, not hidden by the high-K result. These limits do not affect the box-path derivations, but they define the present empirical scope.

VI. Conclusion

KaryoFlow aligns trajectory learning, multistep state identity, and final localization-aware ranking for dense chromosome detection. RF supplies the few-step generative foundation; direct data prediction improves three-seed mAP by 0.002 on both datasets; and the proposal-identity analysis makes DPM++ history mathematically coherent at the retained proposal count. The independent decision contribution, LQCR, improves fixed-checkpoint mAP by 0.0074 and AP90/AP95 by $0.0304/0.0325$ without changing detections before ranking. Together with orthogonal inference compression, these results establish a competitive and interpretable generative detector for chromosome karyotyping.

References

[1] Z. Cai and N. Vasconcelos, “Cascade R-CNN: Delving into high quality object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 6154–6162.

[2] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, “YOLOX: Exceeding YOLO series in 2021,” arXiv:2107.08430, 2021. [Online]. Available: https://arxiv.org/abs/2107.08430

[3] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, “DINO: DETR with improved denoising anchor boxes for end-to-end object detection,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[4] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, “Deformable DETR: Deformable transformers for end-to-end object detection,” in *Int. Conf. Learn. Represent. (ICLR)*, 2021.

[5] S. Chen, P. Sun, Y. Song, and P. Luo, “DiffusionDet: Diffusion model for object detection,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 1980–1989.

[6] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[7] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models,” *Machine Intelligence Research*, 2025.

[8] W. C. Schwartzkopf, A. C. Bovik, and B. L. Evans, “Maximum-likelihood techniques for joint segmentation–classification of multispectral chromosome images,” *IEEE Transactions on Medical Imaging*, vol. 24, no. 12, pp. 1593–1610, 2005.

[9] Y. Qin, J. Wen, H. Zheng, X. Huang, J. Yang, N. Song, Y.-M. Zhu, L. Wu, and G.-Z. Yang, “Varifocal-Net: A chromosome classification approach using deep convolutional networks,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 11, pp. 2569–2581, 2019.

[10] L. Xiao, C. Luo, T. Yu, Y. Luo, M. Wang, F. Yu, Y. Li, C. Tian, and J. Qiao, “DeepACEv2: Automated chromosome enumeration in metaphase cell images using deep convolutional neural networks,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 12, pp. 3920–3932, 2020.

[11] J. Li, S. Zheng, Z. Shui, S. Zhang, L. Yang, Y. Sun, Y. Zhang, H. Li, Y. Ye, P. M. A. van Ooijen, K. Li, and L. Yang, “Masked conditional variational autoencoders for chromosome straightening,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 1, pp. 216–228, 2024.

[12] C.-E. Kuo, J.-Z. Li, J.-J. Tseng, F.-C. Lo, M.-J. Chen, and C.-H. Lu, “ChromosomeNet: Deep learning-based automated chromosome detection in metaphase cell images,” *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 6, pp. 227–236, 2024.

[13] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2015, pp. 91–99.

[14] Y. Xue, G. Bigras, J. Hugh, and N. Ray, “Training convolutional neural networks and compressed sensing end-to-end for microscopy cell detection,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 11, pp. 2632–2641, 2019.

[15] N. O. Loewke, S. Pai, C. Cordeiro, D. Black, B. L. King, C. H. Contag, B. Chen, T. M. Baer, and O. Solgaard, “Automated cell segmentation for quantitative phase microscopy,” *IEEE Transactions on Medical Imaging*, vol. 37, no. 4, pp. 929–940, 2018.

[16] F. Xing, T. C. Cornish, T. D. Bennett, and D. Ghosh, “Bidirectional mapping-based domain adaptation for nucleus detection in cross-modality microscopy images,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 10, pp. 2880–2896, 2021.

[17] E. Baty, C. P. Bridges, and S. Hadfield, “FlowDet: Unifying object detection and generative transport flows,” arXiv:2512.16771, 2025. [Online]. Available: https://arxiv.org/abs/2512.16771

[18] H. Lee, C. Lee, N. Seo, J. S. Lim, and H. Hong, “DeFloMat: Detection with flow matching for stable and efficient generative object localization,” arXiv:2512.22406, 2025. [Online]. Available: https://arxiv.org/abs/2512.22406

[19] X. Chen, Z. Liu, K. Z. Luo, S. Datta, A. Polavaram, Y. Wang *et al.*, “DiffuBox: Refining 3D object detection with point diffusion,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[20] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[21] C. Jones, E. Noutahi, J. Hartford, and C. Eastwood, “Elucidating the design space of flow matching for cellular microscopy,” arXiv:2603.26790, 2026. [Online]. Available: https://arxiv.org/abs/2603.26790

[22] F. Nützel, M. Dombrowski, and B. Kainz, “Flow matching with optimized subclass priors for medical image augmentation,” arXiv:2605.16469, 2026. [Online]. Available: https://arxiv.org/abs/2605.16469

[23] M. Tivnan, J. Teneggi, T.-C. Lee, R. Zhang, K. Boedeker, L. Cai, G. J. Gang, J. Sulam, and J. W. Stayman, “Fourier diffusion models: A method to control MTF and NPS in score-based stochastic image generation,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 9, pp. 3694–3704, 2025.

[24] H. Yang, W. Hua, Z. Xu, and J. Sun, “Domain-generalized discrete diffusion model for cross-domain medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 11, pp. 4131–4142, 2025.

[25] X. Shen, H. Huang, B. Nichyporuk, and T. Arbel, “Improving robustness and reliability in medical image classification with latent-guided diffusion and nested-ensembles,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 12, pp. 4890–4902, 2025.

[26] W. Yang, S. Gao, H. Zhang, H. Yu, M. Xu, P. Chong, W. Zhang, H. Wang, W. Zhang, and A. Qian, “PtbNet: Based on local few-shot classes and small objects to accurately detect PTB,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 12, pp. 4331–4343, 2024.

[27] K. Yan, J. Cai, Y. Zheng, A. P. Harrison, D. Jin, Y. Tang, Y. Tang, L. Huang, J. Xiao, and L. Lu, “Learning from multiple datasets with heterogeneous and partial labels for universal lesion detection in CT,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 10, pp. 2759–2770, 2021.

[28] C. Chen, J. Wang, J. Pan, C. Bian, and Z. Zhang, “GraphSKT: Graph-guided structured knowledge transfer for domain adaptive lesion detection,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 2, pp. 507–518, 2023.

[29] Y. Ren, X. Liu, J. Ge, Z. Liang, X. Xu, L. J. Grimm, J. Go, J. R. Marks, and J. Y. Lo, “Ipsilateral lesion detection refinement for tomosynthesis,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 10, pp. 3080–3090, 2023.

[30] R. Wang, B. Chen, D. Meng, and L. Wang, “Weakly supervised lesion detection from fundus images,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 6, pp. 1501–1512, 2019.

[31] S.-H. Bae and K.-J. Yoon, “Polyp detection via imbalanced learning and discriminative feature learning,” *IEEE Transactions on Medical Imaging*, vol. 34, no. 11, pp. 2379–2393, 2015.

[32] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, “Acquisition of localization confidence for accurate object detection,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 784–799.

[33] X. Li, W. Wang, L. Wu, S. Chen, X. Hu, J. Li, J. Tang, and J. Yang, “Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 21 002–21 012.

[34] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, “VarifocalNet: An IoU-aware dense object detector,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 8514–8523.

[35] A. Mehrtash, W. M. Wells, C. M. Tempany, P. Abolmaesumi, and T. Kapur, “Confidence calibration and predictive uncertainty estimation for deep medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 12, pp. 3868–3878, 2020.

[36] P. Dhariwal and A. Nichol, “Diffusion models beat GANs on image synthesis,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.

[37] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio, “Conditional flow matching: Simulation-free dynamic optimal transport,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[38] A.-A. Pooladian, H. Ben-Hamu, C. Domingo-Enrich, B. Amos, Y. Lipman, and R. T. Q. Chen, “Multisample flow matching: Straightening flows with minibatch couplings,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 28 100–28 127.

[39] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2013, pp. 2292–2300.

[40] H. K. Cheng and A. Schwing, “The curse of conditions: Analyzing and improving optimal transport for conditional flow-based generation,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 15 875–15 884.

[41] South China Normal University, “RST (chromosome20240904) dataset,” Roboflow Universe, 2024, accessed Jul. 2026. [Online]. Available: https://universe.roboflow.com/south-china-normal-university-imqzk/rst

[42] J.-J. Tseng, C.-H. Lu, J.-Z. Li, H.-Y. Lai, M.-H. Chen, F.-Y. Cheng, and C.-E. Kuo, “An open dataset of annotated metaphase cell images for chromosome identification,” *Scientific Data*, vol. 10, p. 104, 2023.

[43] C.-H. Lu, C.-E. Kuo, and J.-J. Tseng, “CIL:54816 - 24 chromosomes object dataset,” Cell Image Library, 2022. [Online]. Available: https://doi.org/10.7295/W9CIL54816

[44] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[45] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature pyramid networks for object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 936–944.

[46] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Int. Conf. Learn. Represent. (ICLR)*, 2019.

[47] ——, “SGDR: Stochastic gradient descent with warm restarts,” in *Int. Conf. Learn. Represent. (ICLR)*, 2017.

[48] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2999–3007.

[49] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, “Generalized intersection over union: A metric and a loss for bounding box regression,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 658–666.

[50] H. W. Kuhn, “The hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.

[51] C. Lyu, W. Zhang, H. Huang, Y. Zhou, Y. Wang, Y. Liu, S. Zhang, and K. Chen, “RTMDet: An empirical study of designing real-time object detectors,” arXiv:2212.07784, 2022.